

Algebra I Exponents and Radicals Cheatsheet

Module 7

Exponent Rules

For $a \neq 0$:

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \quad (b \neq 0)$$

$$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$$

Scientific Notation

$$a \times 10^n$$

where

$$1 \leq |a| < 10$$

Large numbers have positive powers of 10. Small decimals have negative powers of 10.

Square Roots

$$\sqrt{a^2} = |a|$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b} \quad (a \geq 0, b \geq 0)$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad (a \geq 0, b > 0)$$

Simplifying Radicals

Pull out perfect-square factors.

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

$$3\sqrt{5} + 7\sqrt{5} = 10\sqrt{5}$$

Only like radicals combine.

Rational and Irrational Numbers

- Rational numbers can be written as $\frac{a}{b}$ with integers a, b and $b \neq 0$.
- Terminating and repeating decimals are rational.
- Non-repeating, non-terminating decimals are irrational.
- Square roots of non-perfect squares are irrational.

Quick Checks

- Negative exponents move factors across the fraction bar.
- Exponents distribute over multiplication, not addition.
- $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ in general.
- Simplified radicals have no perfect-square factors inside.